Review article: Numerical Methods for Engineering Quantum Computers

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ABSTRACT

In the article by Herrmann et al, “Quantum utility – definition and assessment of a practical quantum advantage,” the quantum computer “Application Readiness Level” (ARL) is defined: (ARL1) “concept with unknown potential,” (ARL2) “beneficial in small idealized systems,” (ARL3) “utility indicated by theory or resource estimations,” (ARL4) “simulated utility demonstration” and (ARL5) “utility demonstration on quantum hardware.” Hermann et al posit that ARL3 has been reached via demonstrations of the Variational Quantum Eigenvalue (VQE) algorithm. In this review article, the latest activities from the computational “digital twin” perspective are reported on in which scientists are working hard to bring practitioners out of the Noisy Intermediate Scale Quantum (NISQ) era into an era of “Quantum supremacy.” In other words, this review article discusses digital twin activities that help to advance from ARL3 to ARL4, or even ARL5. The main categories in this review article are (1) Optimization, Control, Inverse problems, surface codes, and exploratory simulation algorithms for Schrodinger’s equation, and (2) Quantum algorithms for benchmarking quantum computers to include emulator algorithms like Qiskit and the Intel Quantum Simulator. The discussion on emulators will include noise models, state preparation, scalability on high performance computers, and available quantum error correction codes.

1. Introduction

The motivation for this review article is the fact that Moore’s law is no longer applicable [96]. As quoted from Shalf [120], “Moore’s Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the ×86, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore’s Law for the past 50 years

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is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore's Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling."

References one through three in the above quote correspond to the following references respectively: [96], [90], [93].

As further motivation for this review article, in Figure 3 of the article by Shalf [120], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three categories: (a) roadmap for the next ten years, (b) 20 years, and (c) "Decades beyond exascale," "New Models of Computation." For category (c), Shalf [120] refers to the following prospective technology: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorhic, and (v) quantum [37, 38].

Of the "new models of computation" listed in Shalf's article [120], this review article will address the "Numerical Methods for Engineering Quantum Computers" aspects associated with the emerging "quantum computing" paradigm. As outlined by Gamble [41], the development of reliable quantum computers will have a tranformative effect on computer technology. A perfectly working fifty qubit quantum computer can efficiently process 2^{50} pieces of data which is 1024 Terabytes (1 Petabyte)!

That being said, right now we are in the "Noisy Intermediate-Scale Quantum" [21] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

2. Numerical methods for computing solutions to Schrodinger's equation and Numerical methods for optimization and control in which the objective function corresponds to PDE constrained optimization with respect to Schrodinger's PDE equation

In 1982, Feit et al [35] developed a spectral method for determining the eigenvalues and eigenfunctions of the time independent Schrodinger equation:

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2M}\nabla^2\psi + V(x,y)\psi \tag{1}$$

$$\psi(x,y,t) = \sum_{n,j} A_{nj}u_{nj}(x,y)e^{-iE_n t} \tag{2}$$

$$E_n u_{nj} = -\frac{1}{2M}\nabla^2 u_{nj} + V u_{nj}, \tag{3}$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}. \tag{4}$$

In 2002, Bao et al [6] developed a time splitting spectral approximation for the time dependent Schrodinger equation in the "semiclassical" regime:

$$iu_t^\epsilon = -\frac{\epsilon}{2}\Delta u^\epsilon + \frac{1}{\epsilon}V(x)u^\epsilon \quad x \in R^d \quad u^\epsilon(x,0) = u_0^\epsilon(x) \tag{5}$$

Bao et al report results in one and two space dimensions.
In 2004, Zheng et al [142] developed a finite element method for finding the ground state of the lithium atom. In otherwords, Zheng et al found the smallest eigenvalue and the associated eigenfunction for the time independent Schrodinger equation:

$$E\psi = H\psi \tag{6}$$

where the Hamiltonian operator H is defined as

$$H = -\frac{1}{2}(\Delta_1 + \Delta_2 + \Delta_3) - \frac{3}{r_1} - \frac{3}{r_2} - \frac{3}{r_3} + \frac{1}{r_{12}} + \frac{1}{r_{13}} + \frac{1}{r_{23}} \tag{7}$$

$$\Delta_j = \frac{\partial^2}{\partial x_j^2} + \frac{\partial^2}{\partial y_j^2} + \frac{\partial^2}{\partial z_j^2} \quad 1 \le j \le 3 \tag{8}$$

$$r_{jk} = |\vec{r}_j - \vec{r}_k| \quad r_j = |\vec{r}_j| \tag{9}$$

After converting from cartesian coordinate system to spherical coordinate system, the authors were able to reduce the dimension of the problem from 9 to 6.
In Grivopoulos' 2005 Phd thesis [45], methods were developed for the control of quantum systems. We highlight one such control problem discussed in the thesis. The Lindblad equation [92, 83] for open quantum systems is (see 3.7 in [45]):

$$\dot{\rho} = -i[H, \rho] + \sum_n K_n \rho K_n' - \frac{1}{2}\{\sum_n K_n' K_n, \rho\} \tag{10}$$

The Lindblad equation models the interaction of two quantum systems, A and B, with respective Hilbert spaces H_A and H_B. The Hilbert space for the composite space is,

$$H_{AB} = H_A \otimes H_B \tag{11}$$

A general state vector on the composite space is,

$$\psi_{AB} = \psi_A \otimes \psi_B. \tag{12}$$

As stated in [45], "For the open system A (interacting with its environment B), ρ is the state variable analogous to the vector ψ for an isolated quantum system." Expressions for ρ are:

$$\rho = \mathrm{tr}_B \psi_{AB} \psi_{AB}' = \sum_{ij} (\sum_I (\psi_{AB})^*_{iI} (\psi_{AB})_{jI}) e_j e_i' \tag{13}$$

Note that ρ is self-adjoint, positive semi-definite, and $\mathrm{tr}\rho = 1$. As an example of control of open quantum systems, [45], considers a control problem for the laser cooling of atoms and molecules. The Lindblad equation with Hamiltonian control is

$$\dot{\rho} = -i[H_0 + Vu(t), \rho] + L(\rho), \tag{14}$$

where

$$L(\rho) = \sum_{ij} K_{ij} \rho K_{ij}^\dagger - \frac{1}{2}\{K_{ij}^\dagger K_{ij}, \rho\} \tag{15}$$

[45], for the 3×3 case, determines the control, $u(t)$, in (14), that maximizes $\rho_{11}(T)$ given $\rho(0) = \rho_0$.
In the article by Han et al [50], a neural network algorithm was devised for computing the ground state to the time independent Schrodinger equation (6). The Hamiltonian operator corresponds to the Born-Oppenheimer approximation [17] with N electrons and M ions:

$$\vec{r} = (\vec{r}_1, \dots, \vec{r}_N) \qquad \vec{R} = (\vec{R}_1, \dots, \vec{R}_M) \tag{16}$$

$$H(\vec{r}, \vec{R}) = -\frac{1}{2}\sum_{i=1}^N \nabla_i^2 + \sum_{i=1}^N \sum_{j=i+1}^N \frac{1}{|\vec{r}_i - \vec{r}_j|} - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{|\vec{r}_i - \vec{R}_I|} + \sum_{I=1}^M \sum_{J=I+1}^M \frac{Z_I Z_J}{|\vec{R}_I - \vec{R}_J|} \tag{17}$$

It is assumed that the first $N_\uparrow$ electrons are spin-up and the remaining $N - N_\uparrow$ electrons are spin down[39]. The trial wave functions are assumed in [50] to be real valued, anti-symmetric, and have the following form:

$$\psi(\vec{r}, \vec{R}) = S(\vec{r}, \vec{R}) A^\uparrow(\vec{r}^\uparrow) A^\downarrow(\vec{r}^\downarrow) \tag{18}$$

[50] represent S,$A^\uparrow$, $A^\downarrow$ using a deep neural network. The weights and biases of the network are trained in order to minimize:

$$E[\psi] = \frac{\psi^*(\vec{r}, \vec{R}) H \psi(\vec{r}, \vec{R}) d\vec{r}}{\psi^*(\vec{r}, \vec{R}) \psi(\vec{r}, \vec{R}) d\vec{r}}. \tag{19}$$

[50] represent the network parameters at epoch t as,

$$\vec{\Theta}, \tag{20}$$

and the cost function in terms of $\vec{\Theta}$ is given as,

$$E[\psi] \equiv \Omega_{E_{ref}}(\vec{\Theta}). \tag{21}$$

The stochastic gradient descent method with learning rate η is used to minimize (21):

$$\Theta_{t+1} = \Theta_t - \eta \nabla_{\vec{\Theta}} \Omega_{E_{ref}}(\vec{\Theta}) \tag{22}$$

[50] report results within two percent error for H_2 (two electrons), Be_4 (four electrons), LiH_4 (four electrons), He (two electrons), and H_{10} (ten electrons).
[54] developed a similar method as [50] for computing the ground state to the time independent Schrodinger equation (6). Both [54] and [50] express the trial wave functions using a neural network:

$$\psi(\vec{r}, \vec{R}) \approx \psi_{\vec{\Theta}}(\vec{r}) \tag{23}$$

[54] report results within three percent error for H_2 (two electrons), Be_4 (four electrons), LiH_4 (four electrons), and B (five electrons).

1. Günther et al (2021) [49], "Quandary: An open-source C++ package for high-performance optimal control of open quantum systems." the article by Lidar (2019) [83] ("Lecture notes on the theory of open quantum systems") gives the derivation of the model for open quantum systems.
2. Günther and Petersson (2023) [48], "A practical approach to determine minimal quantum gate durations using amplitude-bounded quantum controls."
3. Fei et al (2025) [33], "Switching time optimization for binary quantum optimal control."
4. Fei et al (2023) [34], "Binary control pulse optimization for quantum systems."
5. Kanai et al (2024) [67], "Single-Electron Qubits Based on Quantum Ring States on Solid Neon Surface."
6. Andrade et al (2022) [3], "Engineering an effective three-spin Hamiltonian in trapped-ion systems for applications in quantum simulation."
7. He et al (2024) [53], "Efficient Optimal Control of Open Quantum Systems"
8. Hangleiter et al (2024) [51], "Robustly learning the Hamiltonian dynamics of a superconducting quantum processor."
9. Sun and Galperin (2025) [128], "Control of open quantum systems: The nonequilibrium Green's function perspective."
10. Lin Lin (2025) [84], "Dissipative preparation of many-body quantum states: Toward practical quantum advantage"
11. Kodali and Motamarri (2025) [76], "Finite-element methods for noncollinear magnetism and spin-orbit coupling in real-space pseudopotential density functional theory"
12. Barrera et al (2025) [8], "The Moving Born-Oppenheimer Approximation"
13. Nieminen et al (2023) [103], "Atomistic modeling of a superconductor–transition metal dichalcogenide–superconductor Josephson junction"
14. Mucciolo et al (2025) [97], "Green's Function Methods for Computing Supercurrents in Josephson Junctions"
15. Williams-Young et al (2023) [136], "Distributed memory, GPU accelerated Fock construction for hybrid, Gaussian basis density functional theory"
16. Dat-Thuc Nguyen and Vo Anh Khoa (2025) [100], "An inversion approach to reconstruct the complex potential and intensity function in the Schrödinger evolution equation"
17. Meschede et al (2023) [95], "Quantum scattering of spinless particles in Riemannian manifolds"
18. Atanasov and Dandoloff (2007) [5], "Curvature induced quantum potential on deformed surfaces"
19. Ferrari and Cuoghi (2008) [36], "Schrödinger equation for a particle on a curved surface in an electric and magnetic field"

3. The Schrodinger inverse problem

I update my process for inverse problem with potential reconstructions and NLS equation.

The time-dependent Schrödinger equation appeared in many studies of particle evolution in quantum dynamics of the systems of electrons, atoms, or molecules. The interactions of atom-photon in quantum optics and potential control to achieve a desired quantum state in quantum control problem are two significant applications in engineering. In the meantime, the inverse coefficient problem for Schrödinger equation arose but only a few numbers of works for reconstructing the complex-valued potentials recorded.

Some of the very first publications in inverse problem for Schrödinger equation were to consider and propose an approach for 1D real-valued potentials. One of those research articles shows that a set of real initial data, Dirichlet boundary data, and partial Neumann data can uniquely and stably rebuild the real-valued spatial potential in one dimensional domain. Later, it was enhanced to allow potential reconstructions from less boundary data and extend to discontinuous principal coefficient. In addition, by measuring a single sub-boundary data, one can also reconstruct spatially real-valued potentials. Multiple-coefficient including two time-independent coefficients and potentials reconstructions in the Schrödinger evolution equation have been well studied. Particularly, electric potential and potential's support are uniquely recovered from a single dynamical data.

The magnetic Schrödinger equation has also been a spotlight in the field of inverse problem where two of the first publications in this direction claimed the stable dependence of the compactly supported magnetic field and the electric potential on the Dirichlet-to-Neumann map. The results were based on the uniqueness of modulo a gauge transform and the so-called optics geometric method. Afterward, this Dirichlet-to-Neumann map approach was expanded to a nonlinear quadratic case and to solve the time-dependent potential in a periodic medium of the non-magnetic inverse problem.

Although there is a huge number of publications regarding to real-valued potential in the time-dependent Schrödinger equation for single-coefficients and multi-coefficients inversion approaches, articles for complex-valued potentials are rare. As far as we concern, the year 2019 saw one of the first mathematical publications that ones can determine the complex-valued electric potential via finitely many partial measurements on Neumann boundary within the entire time domain of the wave-function. Later, this approach was proven for the complex electromagnetic potential reconstruction in two-state magnetic Schrödinger system. Moreover, a physics-informed neural networks (PINNs) was designed to access the inversion of the 1D and 2D potential's parameters. This approach was based on the full Dirichlet-to-Neumann map and measurements on the whole boundary. PINNs then was extended for the entire potential reconstruction in saturable nonlinear Schrödinger equations. Another powerful method named Regularized-Newton-GMRES have also been developed for the complex potential in general nonlinear Schrödinger equation.

From that, I think there will be at least 15 papers needed to be add. I will add more details on this, with formula, methods,...

4. Quantum emulators, large scale Quantum emulators, and Quantum algorithms for benchmarking quantum computers

4.1. Quantum Emulators

The development of Quantum Emulators is essential for benchmarking quantum computers since (i) the level of noise in Quantum Emulators can be prescribed exactly thereby enabling practitioners to quantify the effect of noise on a given quantum circuit design, (ii) one can prescribe a given noise model[85] and then compare emulator results with an actual quantum computer in order to characterise the actual noise present in a quantum computer, and (iii) one can isolate quantum algorithmic problems from intrinsic quantum computer noise problems.

A Quantum algorithm consists of (i) state preparation and measurement and (ii) a sequence of (basic) unitary operation(s), and (iii) a quantum error correction code.

4.1.1. state preparation and measurement

1. State Preparation.

Given n qubits, there will be $N = 2^n$ basis states. For example if $n = 3$, the standard basis is

$$\begin{aligned} |000\rangle &= |0\rangle \otimes |0\rangle \otimes |0\rangle \equiv \vec{\mathbf{0}} \\ |001\rangle &= |0\rangle \otimes |0\rangle \otimes |1\rangle \equiv \vec{\mathbf{1}} \\ &\dots \\ |111\rangle &= |1\rangle \otimes |1\rangle \otimes |1\rangle \equiv \vec{\mathbf{7}} \end{aligned}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In general if one wants to prepare the state,

$$|\vec{\psi}\rangle \qquad |\vec{\psi}\rangle \in C^N,$$

on a quantum computer, then one searches for as sparse a unitary matrix U_ψ as possible such that,

$$U_\psi |\vec{\mathbf{0}}\rangle = |\vec{\psi}\rangle,$$

or, equivalently,

$$U_\psi |\vec{\mathbf{0}}\rangle = \frac{e^{i\theta}}{||\psi||} \sum_{i_1 i_2 \dots i_n} \psi_{i_1 i_2 \dots i_n} |i_1 i_2 \dots i_n\rangle.$$

The indices i_k take values in $0, 1$, $\theta \in [0, 2\pi)$, and

$$||\psi||^2 = \sum_{i_1 i_2 \dots i_n} |\psi_{i_1 i_2 \dots i_n}|^2$$

In preparing an n qubit quantum state, if no ancillary qubits are to be used, then the circuit depth is $O(2^n)$ where the circuit depth is the number of "basic" quantum computer operations that cannot be performed in parallel. As reported in [61, 108], the lower and upper bounds on the number of CNOT gates required for state preparation of n qubits, with no additional ancillaries, is $(1/2)2^n$ and $(23/24)2^n$ respectively. As reported in [141], an optimal circuit depth of $O(n)$ can be achieved if the number of ancilla qubits is $O(2^n)$ [129],[115] (Corollary 1.9).

The results in [129, 115, 108] describe a very important trade off. The larger the state preparation circuit depth, the higher the odds that deocoherence ruins the prepared state, on the other hand, the larger the number of ancillary qubits, the larger the quantum computer that must be built. The quantum singular value decomposition of a dense matrix[116] is an example in which the trade-off between circuit depth and number of ancillar qubits must be considered.

2. Measurement.

On a n qubit system, if one wants to make a measurement of the state $|\psi\rangle$,

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |\boldsymbol{i}\rangle, \tag{24}$$

with respect to the standard basis then one can proceed as follows:

1. Define a complete set of projection operators P_j, $j = 0, \dots, J-1$; e.g.,

   $$P_j = |\boldsymbol{j}\rangle\langle\boldsymbol{j}| \qquad J = 2^n. \tag{25}$$

   It is required that,

   $$\sum_{j=0}^{J-1} P_j = \mathrm{I}.$$

2. The probability of getting an outcome j on measuring the observable P_j is given by,

   $$p(j) = \langle\psi| P_j |\psi\rangle,$$

   and the normalized state after the measurement is given by,

   $$|\psi'\rangle = \frac{P_j |\psi\rangle}{\sqrt{\langle\psi| P_j |\psi\rangle}}. \tag{26}$$

In order to make a measurement in an arbitrary basis[126], then one can proceed as follows:

1. create a unitary matrix $\boldsymbol{U}$ with the arbitrary basis vectors,

   $$|\boldsymbol{u}_0\rangle, \ldots, |\boldsymbol{u}_{2^n-1}\rangle,$$

   as columns.
2. Apply $\boldsymbol{U}^\dagger$:

   $$|\tilde{\psi}\rangle = U^\dagger |\psi\rangle.$$

   The coefficients with respect to the new basis are now the coefficients of $|\tilde{\psi}\rangle$ with respect to the standard basis:

   $$|\tilde{\psi}\rangle = \sum_{i=0}^{2^n-1} \beta_i |\boldsymbol{i}\rangle \qquad |\psi\rangle = \sum_{i=0}^{2^n-1} \beta_i |\boldsymbol{u_i}\rangle$$

3. Measure $|\tilde{\psi}\rangle$ using the same steps as if one was making a measurement with respect to the standard basis, except that one must multiply the measured state (26) by $\boldsymbol{U}$:

   $$|\psi'\rangle = \frac{U P_j |\tilde{\psi}\rangle}{\sqrt{\langle\tilde{\psi}| P_j |\tilde{\psi}\rangle}}. \tag{27}$$

Another way to think about making a measurement in an arbitrary basis is to first define the projectors,

$$P_j \equiv |\boldsymbol{u_j}\rangle \langle\boldsymbol{u_j}| \qquad j = 0, \ldots, 2^n - 1. \tag{28}$$

Then, one measures $|\psi\rangle$ with respect to the projectors in (28) which is equivalent to measuring $|\tilde{\psi}\rangle$ with respect to the standard basis and then multiplying the result by U.
One can define an “Observable” as follows,

$$A = \lambda_0 P_0 + \lambda_1 P_1 + \ldots + \lambda_{2^n-1} P_{2^n-1} \tag{29}$$

where the eigenvalues of A are λ_j and the corresponding eigenvectors of A are $|\boldsymbol{u_j}\rangle$. As quoted from [126], “when we measure an observable A in a state $|\psi\rangle$, each measurement yields exactly one of the eigenvalues λ_j. The probability of obtaining λ_j is,

$$p(j) = \langle\psi| P_j |\psi\rangle. \tag{30}$$

Upon measurement, the state “collapses” to a new state

$$\frac{P_j |\psi\rangle}{\sqrt{\langle\psi| P_j |\psi\rangle}}. \tag{31}$$

The denominator is the normalization factor to account for the unit normal requirement of a quantum state.”
A very fundamental benchmark test is to repeatedly (a) prepare a given state, and (b) measure the state with respect to an observable A. In what follows, many quantum emulators are compared and contrasted with respect to their (i) state preparation algorithm, (ii) noise modeling, (iii) parallel capability, (iv) vectorizable capability, and (v) ability to test quantum error correction algorithms for a given noise level.
The IBM Qiskit emulator[63] uses the algorithm described in Iten et al[61] for state preparation. First it was reported in [61] that for the state preparation of n qubits, a lower bound on the number of CNOT gates required is $2^n/2$ and an upper bound is $(23/24)2^n$[108]. In Appendix A5 of [61], optimized state preparation algorithms were described for the state preparation of $n = 2$ and $n = 3$ qubits[143]; for $n \geq 4$, the algorithm from [108] is used.

The following state preparation algorithms have been cited with respect to the Intel Quantum simulator [146], [46], [122], [121]. Furthermore, the following noise models are available on the Intel Quantum Simulator [46]: [9], [118].

4.1.2. Quantum Emulators for converting arbitrary unitary operations into basic quantum gates available on quantum computers.

In the Pennylane quantum emulator, the following article[133] discusses how a given unitary matrix is converted, via the global Cartan decomposition[134], into basic quantum gates. The circuit depth for the Cartan decomposition has been found to be $\frac{21}{16}4^n$ for n qubits[91].

In [133], the following other important articles[121, 79, 81, 26] are cited for carrying out arbitrary unitary operations.

The Qiskit quantum emulator[106] references the following article for synthesizing general, linear, reversible circuits[105].

4.1.3. Quantum Emulators for testing quantum error correction codes

Shor[124] demonstrates that quantum computation, though highly susceptible to decoherence, can be made robust through explicit quantum error correction without destroying superposition or entanglement. Assuming independent decoherence across qubits, the paper presents a concrete nine-qubit encoding that maps each logical qubit into three entangled triplets, allowing perfect recovery of the original quantum state provided at most one qubit per block decoheres. Quantitatively, if each physical qubit decoheres with probability p, the probability that a nine-qubit block fails scales as $O(p^2)$, with decoding success approximately $1 - 36p^2$ for small p, representing a quadratic suppression of errors relative to naive storage. For k logical qubits, the probability of successful recovery behaves as $(1 - 36p^2)^k$, yielding improved storage fidelity whenever $p < 1/36$. The scheme also enables a watchdog effect, in which periodic error detection and correction at intervals much shorter than the decoherence time can, in principle, preserve quantum information indefinitely, limited only by imperfections in the correcting operations. Although the encoding incurs a ninefold qubit overhead and introduces additional gate errors, this work provides the first constructive computational demonstration that quantum error rates can be reduced algorithmically, laying foundational groundwork for fault-tolerant quantum computation.

The following article discusses a more modern error correction code[80], which finds the optimal quantum error correction code, given the constraints proved by Eastin and Knill[31].

The Qiskit quantum emulator documentation[107] gives an example for reducing the probability of a physical qubit undergoing an erroneous bit flip by replacing the physical qubit with a “3-qubit” logical qubit plus two additional ancilla qubits. The IBM sample code will not correct all errors; the IBM code only successfully corrects the logical quibit if only one of its’ three physical qubits undergoes an inadvertant IIX, IXI, or XII command.

4.2. Large scale Quantum Emulators

For an n qubit system, the required storage is (assuming double precision) is $S = (2)(8)2^n$ bytes. For example, if $n = 40$, then $S \approx 18$ terabytes (TB). If data was stored on a 16 TB harddrive, then the time it takes to execute a single quantum gate on any given qubit would take about 20 hours. If the data was stored in RAM, then the time would be only a few minutes in order to execute a one-qubit quantum gate. If $n = 50$, then it would be fastest to implement quantum algorithms on many comptuter nodes. The popular emulators that exploit large scale computer systems are “Atlas”[138], “cuQuantum”[11], “PennyLane-Lightning MPI”[68], and “Massively Parallel Quantum computer simulator”[27].

In the “Atlas”[138] architectural model, there are 2^G nodes, each node contains 2^R GPU cards, and each GPU card stores 2^L (complex)amplitudes. So if there are $n = L + R + G$ qubits, then the state can be written as,

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i \, |i\rangle \, .$$

If we expand i in its binary digits, then, $|i\rangle \equiv |b_0 b_1 \dots b_{L-1} b_L b_{L+1} \dots b_{L+R+G-1}\rangle$. The convention here is that the binary expansion is from least significant to most significant binary digit. The binary representation of the GPU card id containing the amplitude of $|i\rangle$ is then $b_L b_{L+1} \dots b_{L+R+G-1}$. The card id’s are labeled $0, \dots, 2^{R+G} - 1$.

4.2.1. State Preparation Benchmark

In this test, we compare the cost for each of the following four high performance emulators, “Atlas”[138], “cuQuantum”[11], “PennyLane-Lightning MPI”[68], and “Massively Parallel Quantum computer simulator”[27] to do random state preparation on n qubits. In order to do a fair test, first we generate the appropriate quantum circuit from “QISKIT”[63] and then write the circuit to a data file. It is expected that the depth of the circuit will be proportional to 2^n. Then we import the circuit into each of the high performance emulators, and then analyze the results.

4.2.2. Singular Value Decomposition Benchmark

4.3. Quantum Algorithms

1. (IBM) “Quantum Computing with Qiskit” (2024) [63]
2. Wille et al (2018) [135], “Computer-aided design for quantum computation.”
3. “Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits” (2020) [46]
4. “PennyLane-Lightning MPI: A massively scalable quantum circuit simulator based on distributed computing in CPU clusters” (2025) [68]
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A. My Appendix

Appendix sections are coded under `\appendix`.

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